Analysis of Algorithms
Assignment 01

1. Code Euclid's Algorithms from the textbook.
2. Code Consecutive integer checking algorithm from the textbook.
3. Code the Middle-school procedure algorithm from the textbook (You will need to code the Sieve of Eratosthenes algorithm too)
4. Find the way C++ uses to *push* and *pop* elements to and from a list to behave like a stack. Create an example that showcases the idea.
5. Create a class that encapsulates the stack behavior describe in point 4 (Ex: MyStack)
6. Find the way C++ uses to *queue* and *dequeue* elements to and from a list to behave like a queue. Create an example that showcases the idea.
7. Create a class that encapsulates the stack behavior describe in point 6 (Ex: MyQueue)
8. Explore which alternatives does C++ Standard Library provide to deal with stacks and queues. Provide examples.
9. Create your own Binary Tree data structure and test the insertion of elements. Insertion will need its own algorithm, research on how to implement it.

Submission

All submissions are done via the Moodle platform. Make sure to upload within the submission period.